Negative exponents

1 Express the following using only positive exponents:

a p^{-2}
b a^{-8}
c $3x^{-2}$
d $10x^{-6}$
e $4a^{-2}$
f $4y^{-3}$
g $\dfrac{1}{a^{-n}}$
h $\dfrac{a^{-n}}{b^{-m}}$
i $p^{-2}q^{3}$
j $8p^{-3}$
k $m^{-5}n^{-4}p^{4}$

2 Express the following using only negative exponents:

a $\dfrac{1}{u^5}$
b $\dfrac{10}{p^8q^5}$

3 Write an expression for the reciprocal of a^3.

4 Find the value of n on the following equations:

a $\dfrac{1}{36} = 6^n$
b $\dfrac{1}{8} = 2^n$

5 Determine the missing exponent so that following equation is always true.

$$\left(a^3b^{-5}\right)^{\square} = a^{-15}b^{25}$$

6 Simplify the following, expressing your answers using positive exponents:

a $4y^{-5} \cdot 3y^{-3}$
b $(6m)^2 \cdot m^{-6}$
c $2y^3 \cdot \left(-2y^{-7}\right)$
d $6y^7 \cdot 2y^{-5} \cdot 5y^3$
e $5p^3q^{-2} \cdot 9p^{-3}q^5$
f $5p^7q^{-7} \cdot 5p^{-7}q^3$
g $8p^5q^{-5} \cdot 4p^{-4}q^3$
h $2p^4q^{-2} \cdot 5p^{-4}q^{-5}$
i $5p^{-6}q^{-9} \cdot \left(-6p^{-4}q^{-3}\right)$
j $\left(w^4\right)^{-5}$
k $\left(w^{-9}\right)^5$
l $\left(5^2\right)^{-p}$
m $\left(4y^4\right)^{-4}$
n $\left(2y^{-5}\right)^4$
o $\left(-4u^{-4}\right)^3$
p $\left(-u^3\right)^{-4}$
q $\left(4m^{-10}\right)^4$
r $\left(4m^{-8}\right)^{-3}$
s $(4m)^{-3}$
t $\left(5y^3\right)^{-3}$
u $\left(4p^8q^4\right)^{-2}$
v $\left(\dfrac{4}{5}u^{-3}\right)^4$

w $\left(\frac{y}{2}\right)^{-5}$

7 Assuming that all variables are non-zero, simplify the following. Express your answers using positive exponents.

a $\frac{12x^3}{4x^7}$

b $\frac{6x^3}{2x^{-3}}$

c $\frac{25x^{-7}}{5x^{-4}}$

d $\left(\frac{a}{b}\right)^{-5}$

e $\left(\frac{a^3}{b^3}\right)^{-5}$

f $\left(\frac{a^2}{b^5}\right)^{-1}$

g $\left(\frac{3a^9}{b^9}\right)^{-2}$

h $\frac{-4x^2}{2x^{-5}}$

i $\frac{-6x^{-7}}{-2x^{-4}}$

j $\frac{z^{-2}x^{-5}}{y^{-7}}$

k $\frac{\left(6y^3z^2\right)^{-4}}{\left(6y^3z^2\right)^{-5}}$

l $\frac{5p^5q^{-4}}{40p^5q^6}$

m $\left(\frac{2y^3x^0}{z^2}\right)^{-4}$

n $\frac{x^2y^{-3}}{x^{-5}y^2}$

o $\frac{\left(m^{-3}\right)^{-1}\cdot\left(m^4\right)^{-3}}{m^3\cdot m^4}$

p $\left(\frac{m^7}{m^{-10}}\right)^2\cdot\left(\frac{m^5}{m^2}\right)^{-3}$

q $\frac{a^2\cdot a^{-5}\cdot b^{-2}}{\left(a\cdot a^{-5}\cdot b^2\right)^2}$